Emissions caused by data centers

Abstract

Data centers worldwide tend to be energy intensive. Servers, storage, and network equipment in data centers consume a lot of power during their functioning. The energy consumed by data centers has tripled from 2018 to 2024.[1] As time progresses, with the increase in adoption and usage of cloud services and artificial intelligence (AI), energy consumption by data centers is projected to rise rapidly.

78% of data centers in the US still operate on power generated from fossil fuels such as coal.[2] Fossil fuels do not burn cleanly. They generate poisonous greenhouse gases that are detrimental to the atmosphere and the environment we live and breathe in.

As data center usage increases, greenhouse gas emissions generated from them are projected to rise significantly, putting an additional burden on the planet.

In short:

1. Data centers consume a lot of power.
2. The power used by data centers is mostly derived from fossil fuels.
3. Data centers are responsible for significant greenhouse gas emissions.

This white paper discusses the problem and proposes a few solutions that can

1. reduce greenhouse gas emissions from data centers.
2. provide a sustainable future.

Acronyms and abbreviations

Acronyms and abbreviations used in this white paper:

1. DC: Data Center
2. AWS: Amazon Web Services
3. AI: Artificial Intelligence
4. CO_2: Carbon dioxide
5. CH_4: Methane
6. N2O: Nitrous oxide
7. IEA: International Energy Agency

[1] "MIT Technology Review," Dec 13, 2024. AI's emissions are about to skyrocket even further.
[2] "MIT Technology Review," Dec 13, 2024. AI's emissions are about to skyrocket even further.

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1. Introduction

Businesses across the globe employ data centers to host their applications today.
A typical data center is a large warehouse filled with rows and rows of servers, storage, and network equipment running round the clock. Businesses run a vast array of software applications on these servers for their operations and customers.

Data centers require significant power for their daily operation. All equipment in a data center

1. consumes power to function.
2. generates enormous heat during its operation.
3. requires water for cooling.

An increase in adoption of cloud services and AI by businesses in recent years has fueled an even further increase in power consumed by data centers.

Power that data centers require is still generated from fossil fuels such as coal. As a result, they emit poisonous greenhouse gases such as carbon dioxide and methane. Greenhouse gases are harmful to the environment and affect life on the planet.

2. Problem Statement

Businesses across the globe host their applications in data centers. These data centers can be:

1. owned and operated by businesses themselves or

2. hosted and managed by third-party service providers on the cloud (e.g., Amazon Web Services (AWS), Microsoft Azure, etc.).

A typical data center is a large warehouse filled with rows and rows of servers, storage, and network equipment running round the clock. Businesses run a vast array of software applications on these servers to:

1. serve their internal operation and
2. provide services to their customers.

All equipment in a data center:

4. consumes power to function.
5. generates enormous heat during its operation.
6. requires water for cooling.

Data centers require significant power for their daily operation. They consume power and water in the below areas:

1. construction of the data center.
2. operation of equipment (servers, storage, and network) in the data center.
3. cooling the equipment.

Here are a few relevant statistics:

1. Data centers worldwide produce up to 1.5% of global greenhouse gases. Data centers emit a lot of greenhouse gases, especially CO_2. As an example, storing and managing just 100 gigabytes of data creates a carbon footprint of 0.2 tons of carbon dioxide (CO_2).[3]
2. Data center emissions are on par with the airline industry.[4] See bar graph below comparing data center emissions to the airline industry.
3. About 4.59% of all the energy consumed in the US goes toward data centers, a figure that's doubled since 2018.[5]

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[3] Arxiv Environmental Burden of United States Data Centers in the Artificial Intelligence Era
[4] Climatiq Measuring greenhouse gas emissions in data centers: the environmental impact of cloud computing
[5] "MIT Technology Review," Dec 13, 2024. AI's emissions are about to skyrocket even further.

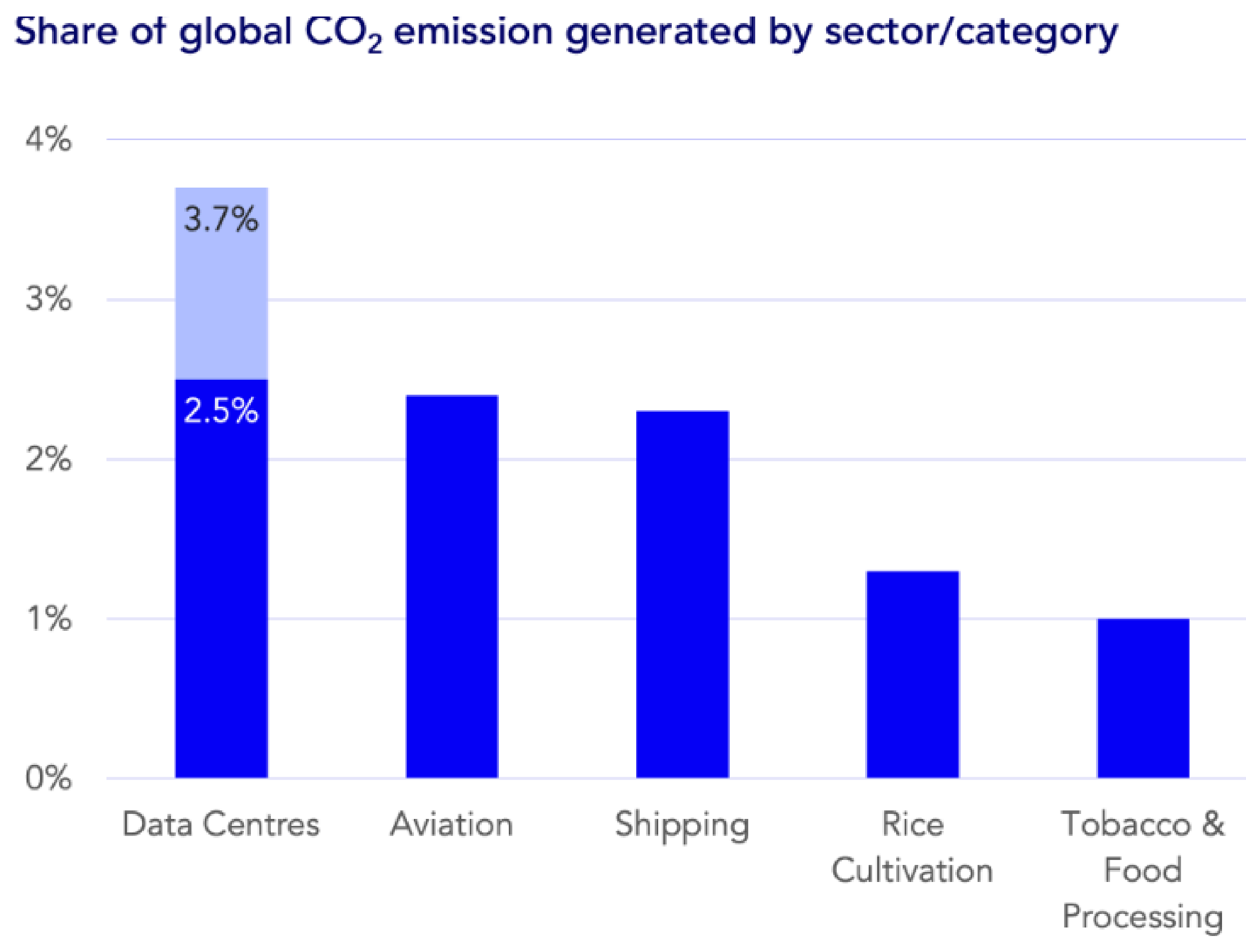


Figure 1: **Share of global CO_2 emission generated by sector/category.**
Source: https://www.climatiq.io/blog/measure-greenhouse-gas-emissions-carbon-data-centres-cloud-computing

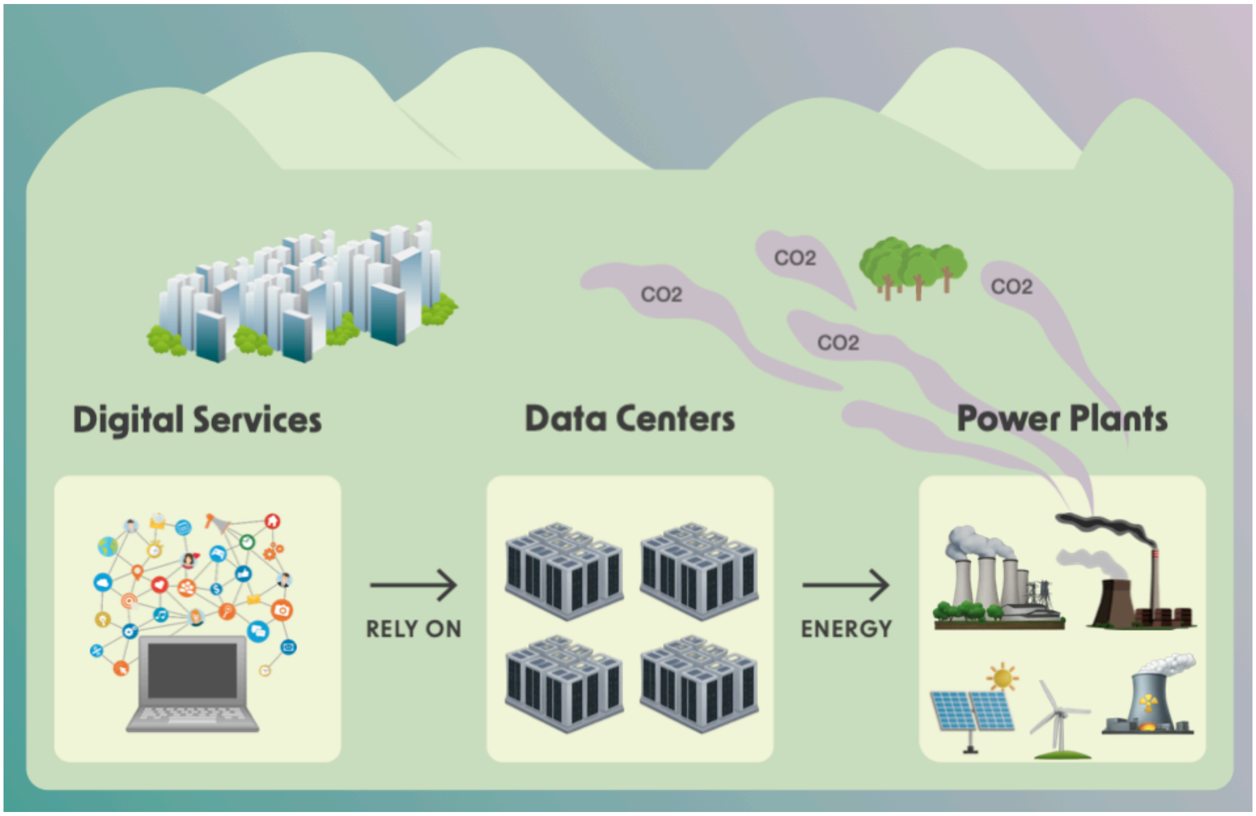


Figure 2:**The chain of digital services, data centers, and CO_2 emissions.**
Source: Data Centers in the Artificial Intelligence Era,
https://arxiv.org/html/2411.09786v1#:~:text=Our%20findings%20reveal%20that%20data,of%20US%20emissions%20in%202023

An increase in adoption of cloud services and AI by businesses in recent years has fueled an even further increase in power consumed by data centers.

As of 2024, about 78% of data centers in the USA still use energy derived from fossil-based fuels such as coal. Such sources result in the emission of the below greenhouse gases:

1. Carbon dioxide (CO_2):
2. Methane (CH_4)
3. Nitrous oxide (N2O)
4. Fluorinated gases

Greenhouse gases are harmful to the environment and affect life on the planet.

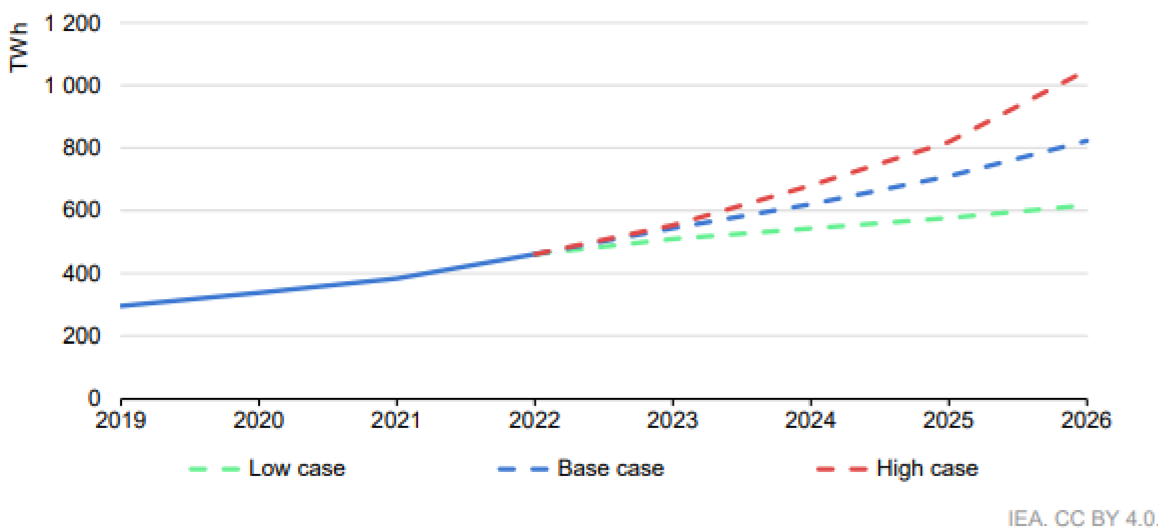
As time progresses, cloud services and AI adoption are set to increase even further. As a result:

1. power consumed by data centers
2. greenhouse gas emissions

are projected to increase rapidly.

The below graph shows electricity consumed by data centers between 2019 and 2026. [6]

Global electricity demand from data centres, AI, and cryptocurrencies, 2019-2026



Notes: Includes traditional data centres, dedicated AI data centres, and cryptocurrency consumption; excludes demand from data transmission networks. The base case scenario has been used in the overall forecast in this report. Low and high case scenarios reflect the uncertainties in the pace of deployment and efficiency gains amid future technological developments.

Sources: Joule (2023), de Vries, The growing energy footprint of AI; CCRI Indices (carbon-ratings.com); The Guardian, Use of AI to reduce data centre energy use; Motors in data centres; The Royal Society, The future of computing beyond Moore's Law; Ireland Central Statistics Office, Data Centres electricity consumption 2022; and Danish Energy Agency, Denmark's energy and climate outlook 2018.

Figure 3: **Global electricity demand from data centers, AI, and cryptocurrencies, 2019-2026**

Source: International Energy Agency, https://www.sustainabilitybynumbers.com/p/ai-energy-demand

The graph below shows the growth in annual greenhouse gas emissions from 1970 to 2023.[7]

[6] Statista Estimated electricity consumption of data centers compared to selected countries.

[7] Statista Energy and Environment: Emissions

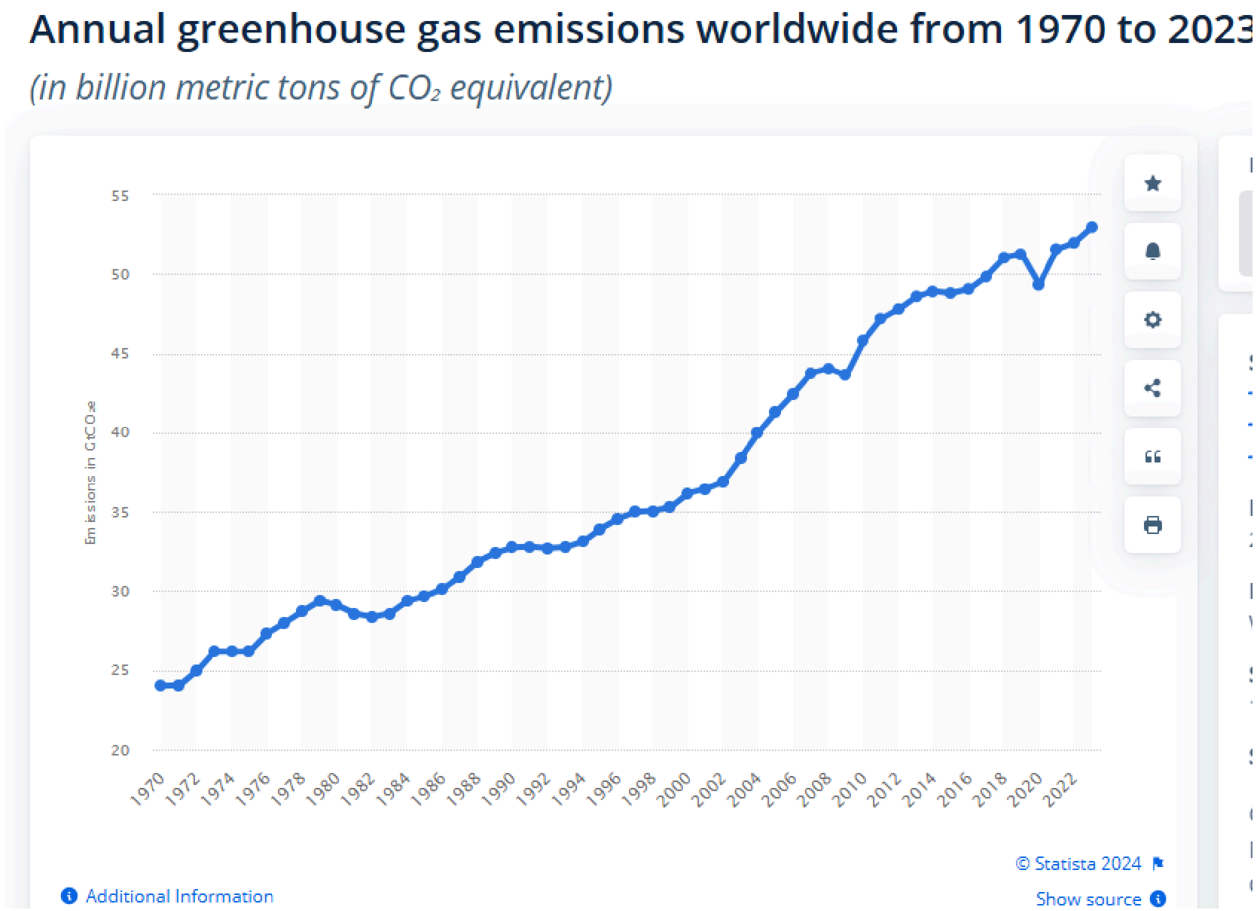


Figure 4: **Annual greenhouse gas emissions worldwide from 1970 to 2023**
Source: Statista, https://www.statista.com/statistics/1285502/annual-global-greenhouse-gas-emissions/#:~:text=In%202023%2C%20global%20greenhouse%20gas.in%20GHG%20emissions%20since%201990.

The global AI market is projected to reach US$184.00bn in 2024.
The market size is expected to show an annual growth rate (CAGR 2024-2030) of 28.46%, resulting in a market volume of US$826.76bn by 2030. The graph below shows the growth trend in AI.

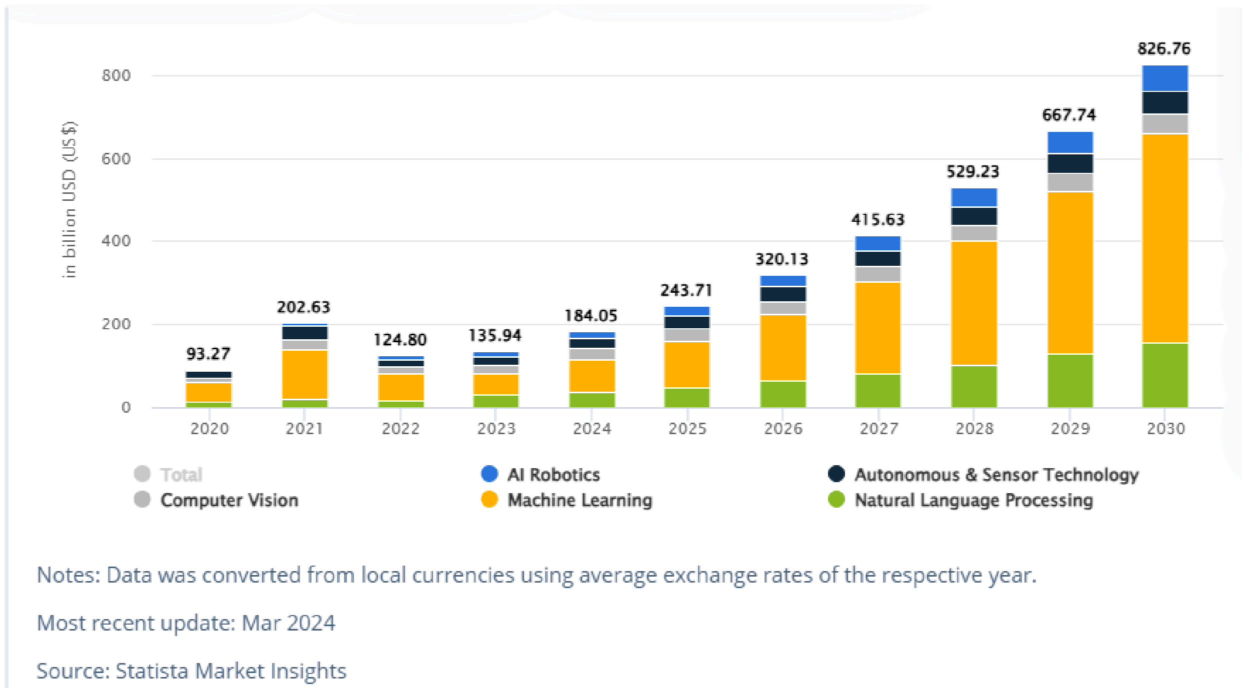


Figure 5: **Artificial Intelligence Market growth from 2020 to 2030**
Source: Statista Market Insights,
https://www.statista.com/outlook/tmo/artificial-intelligence/worldwide#market-size

Businesses need to realize the enormity of this problem, its impact on the greater environment, and act on it with urgency. They need to measure, analyze, and provide a tangible solution to significantly reducing the greenhouse gases emitted from the data centers they operate.

3. Solution

Businesses need to look at reducing carbon emissions as an “urgent” priority for their organization.

Here are some ways to reduce carbon emissions:[8]

1. Use renewable energy.
   Data centers can use renewable energy sources like solar, wind, and hydropower to reduce their reliance on fossil fuels.
   A significant number of large data centers (78% in the USA) still continue to rely on traditional fossil fuel-based energy resources such as coal.
   However, switching to renewable energy sources like
   a. Solar
   b. Wind
   c. Hydropower

[8] Hitachi Vantara: Decarbonizing the data center for dummies

brings significant reduction in greenhouse gas emissions.

2. Use energy-efficient hardware.
   Prioritize servers, hard drives, and other components with high energy efficiency ratings. Manufacturers of servers, storage, and network equipment continue to innovate and make changes to their design to be energy efficient and low-power consuming. Based on the energy efficiency ratings provided by manufacturers, if you upgrade to and switch to newer versions of hardware, data centers can benefit from the power-saving and auto-cooling. features that the newer hardware brings along with it. Old server models and hardware tend to be power hungry. The equipment in data centers typically needs a refresh to newer hardware every 3 to 4 years.

3. Reduce hardware footprint in the data center.
   Use virtualization instead of physical hardware.Virtualization is a modern technology that allows for efficient utilization of physical hardware. This technology creates a virtualization layer on top of physical hardware and lets multiple users share and use a slice of the underlying hardware simultaneously. Such shared use of a single physical server or hardware by multiple users results in a significant reduction in terms of hardware footprint. This also results in a reduction of power consumption in the data center.

4. Compress data to reduce storage requirements.
   When data is compressed and stored, the storage requirements in the data center reduce drastically compared to storing data as is. As a result, power consumption by storage devices reduces significantly when data is compressed and stored.

5. Turn off services when not in use.
   Turn off services automatically when not in use and turn on services automatically when required. In data centers, servers and services continue to run non-stop even if no user is accessing these services, resulting in wasted power usage. Setting up ways to automatically turn services off during quiet periods and turning them back on automatically when required helps save power significantly.

6. Extend the lifecycle of hardware.

7. Use fuel cells
   Some data centers are exploring innovative technologies such as fuel cells, which can generate electricity through a chemical reaction.

8. Carbon offset

Businesses can participate in a carbon offset program. It is a way to compensate for greenhouse gas emissions by funding projects that reduce or store carbon.

People or groups buy carbon offset certificates, which are linked to activities that reduce carbon dioxide (CO2) emissions, such as planting trees. The certificates "offset" the buyer's CO2 emissions with an equal amount of CO2 reductions elsewhere.

4. Conclusion

There is a significant increase in the adoption and use of cloud services and AI by businesses worldwide in recent years. This has led to

1. a corresponding increase in energy consumption in data centers that provide these services.
2. greenhouse gas emissions by data centers that use power mostly derived from fossil fuels.

A number of strategies are available to reduce power consumption and greenhouse gas emissions:

1. Use renewable energy.
2. Use energy-efficient hardware.
3. Reduce hardware footprint in the data center.
4. Compress data to reduce storage requirements.
5. Turn off services when not in use.
6. Extend the lifecycle of hardware.
7. Use fuel cells

Businesses need to evaluate and implement one or more strategies on priority to keep

1. power consumption in data centers under control
2. greenhouse gas emissions at a manageable level

5. References

a. https://www.technologyreview.com/2024/12/13/1108719/ais-emissions-are-about-to-skyrocket-even-further/
b. https://www.climatiq.io/blog/measure-greenhouse-gas-emissions-carbon-data-centres-cloud-computing
c. https://arxiv.org/html/2411.09786v1#:~:text=Our%20findings%20reveal%20that%20data,of%20US%20emissions%20in%202023
d. https://www.statista.com/outlook/tmo/artificial-intelligence/worldwide#market-size
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